

Home Lab on a Shoestring

A real server, your own services, and a network you actually understand — starting with **what's already in your closet**. The skills are the same ones a junior sysadmin gets paid for. Cost to start: **\$0**.

START WITH WHAT YOU OWN

- **The old laptop or desktop** in the closet. This is your server. Almost any machine from the last decade runs Linux fine.
- **A USB stick** (8GB+). Your install media.
- **Your home router**. You already pay for it. Log in tonight and learn what it can do.
- **Your phone**. A client, a scanner, a remote control.

FIVE \$0 WINS, IN ORDER

- 1 **Reserve one address**. On the router, give your server a fixed DHCP reservation. One setting, and you stop chasing IPs.
- 2 **Block ads for the whole house**. Run Pi-hole (or AdGuard Home) in Docker and point the network at it. You feel it the same day.
- 3 **Keys, not passwords**. SSH keys to the box, then Tailscale (free tier) to reach it from anywhere — without opening a single port.
- 4 **One compose file per service**. Learn Docker Compose once and every app after it is a few lines.
- 5 **Jellyfin for your media**. Point it at one folder. Grow it later.

ORDER OF OPERATIONS

- ☐ Wipe and install Debian or Ubuntu Server.
- ☐ Update it: **apt update && apt full-upgrade**.
- ☐ SSH in from another machine; confirm it survives a reboot.
- ☐ Router: DHCP reservation for the server.
- ☐ Docker + Compose installed.
- ☐ Pi-hole up; set the network's DNS to the server.
- ☐ Tailscale up on server and phone; reach it off your Wi-Fi.
- ☐ Jellyfin up; play one file end to end.
- ☐ Write it all into one **notes.md**. Config-log notes are a gift to future you.

THE ONE THING TO RESPECT

Never expose a service straight to the internet — reach it through the VPN. Change every default password. Keep one backup of your config folder. That is the entire security baseline, and it is enough to start.

WHEN YOU'RE READY TO SPEND

Don't buy anything until the free build is running. Then, in this order:

- **A used small-form-factor PC** (Dell or Lenovo, roughly \$40–60). Real CPU, real RAM, quiet.
- **A Raspberry Pi kit** (about \$80) when you want low power and always-on.
- **A cheap managed switch** the day you outgrow one network port.

Path: **\$0 today, under \$100 when you're ready**.